# Jordan Hill — Curriculum Vitae

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## Research profile

I am a PhD candidate in Artificial Intelligence at the University of Western Australia asking whether AI systems could share our *kairotic positionality*, the position within an unfolding situation from which a fitting moment becomes available, and what that would mean for alignment, morality and our own nature. I pursue it through philosophical inquiry and empirical exploration, building the systems I study: I am developing **Kairos**, a novel neural architecture in which an incoming signal, its learned representations and a language model advance at different rates while remaining responsive to one another. Its alignment premise is that systems sharing our relation to time would be less alien to us, and therefore more predictable. The project's linguistic strand, accepted for presentation at ALS 2026, separates reduced latency from well-timed response through prosody, pauses, turn-taking and repair. Supervision across philosophy, computer science and linguistics, with earlier training in Ancient Greek, classical philosophy and medieval studies, lets me work from primary sources and running code alike. I build and run my own research infrastructure, from gradient-checked reference implementations to an agent-native experiment system on two GPU clusters.

## Education

**Doctor of Philosophy (Artificial Intelligence), University of Western Australia** | 2026–present. Thesis: *Kairos: A Series of Unresolved Analogies Between Two Natures*; expected submission August 2029. Supervisors: Associate Professor Nin Kirkham (Philosophy; principal), Dr Wei Liu (Computer Science), Associate Professor Celeste Rodriguez Louro (Linguistics). Australian Government Research Training Program Scholarship.

**Graduate Certificate in Data Science, RMIT University** | 2024. GPA 3.8 in machine learning and data analysis in Python and R.

**Master of Arts in the Liberal Arts, St John's College, Annapolis, Maryland** | 2023. GPA 3.95, specialising in Ancient Greek through two years of intensive language study, conducted entirely through seminar and close reading of primary texts. Oxford-style tutorials translating Sappho's lyric fragments (with Katherine Kretler) and Xenophon's *Memorabilia*, each defending independent translations in dialogue with the tutor. Ramsay Postgraduate Scholarship.

**Bachelor of Arts (Honours), First Class, History, University of Western Australia** | 2019. Thesis: *Transformation and Identity in the Magic of the Mabinogi* (15,000 words), supervised by Associate Professor Kirk Essary, which earned the highest mark in its area and the Philippa Maddern Memorial Prize.

**Bachelor of Arts, University of Western Australia** | 2018. Double major in English and Cultural Studies, and History.

**Certificate IV in Training and Assessment, North Metropolitan TAFE** | 2025.

## Research

### Kairos: a novel neural architecture for signal processing | 2026–present

*Doctoral research and Prosody Labs; unpublished.*

- **Architecture.** Learned encoders and decoders let a sequence-model core (Gated DeltaNet, attention or a hybrid) run at its own rate, independent of the rates at which signals enter and leave the model, so one architecture serves raw audio, text and both together.
- **Proof-of-concept audio reconstruction.** In the Kairos audio playground, encodes raw 8 kHz audio into learned representations, carries them through a fixed-rate bottleneck in which a Gated DeltaNet core runs at 1 kHz (8× fewer positions), and reconstructs the waveform at approximately 23 dB signal-to-noise ratio on a recorded clip.
- **Full-duplex conversation.** Extends Moshi (Défossez et al., 2024), the basis of NVIDIA's PersonaPlex. Moshi hears and speaks through Mimi, a separately trained codec that quantises 24 kHz speech into eight discrete codebooks at a fixed 12.5 Hz; Kairos instead learns continuous representations from raw waveform end to end, on audio and text clocks that advance independently, scaling the playground's reconstruction to the full English LibriVox corpus. Demonstrated end-to-end training at 0.8 billion parameters (Qwen3.5-0.8B) on four NVIDIA H100s over 12,000 updates, with promising early traction; the comparison against a Moshi-recipe baseline is still to come.
- **Byte-level language modelling from signals.** Combines text encoded as a signal for Kairos with byte-level multi-token prediction, extending the 8-byte horizon of Gloeckle et al. (2024) to 32 bytes, scaling to 64. The 256-symbol byte vocabulary keeps many prediction heads cheap where token vocabularies make them a memory bottleneck, and the longer horizon trains a tokenizer-free model towards word-scale structure. Trained on four AMD MI210s and NVIDIA H100s and evaluated on the CORE benchmark suite, alongside an original recurrent token mixer.
- **Evidence-led redesign.** Earlier versions, tested inside recurrent and attention-based language models of 120–150 million parameters, redirected the architecture towards variable-rate processing.

### Research infrastructure | Prosody Labs, 2026–present

- **Execute.** An agent-native experiment execution system for Slurm: every run is a commit-pinned, auditable record queued through git, by hand or by AI coding agents, on UWA's Kaya HPC cluster and a dedicated workstation, with guardrails for unattended use. Over 1,100 runs since July 2026.
- **Verified training.** A gradient-checked reference implementation and a PyTorch training system validated against it keep large-scale results traceable to verified mathematics.
- **Cross-vendor kernels and training.** Custom Triton kernels for both NVIDIA CUDA and AMD ROCm, qualified against the reference on RTX 4090 and MI210 GPUs, with FSDP and DDP training across H100 and MI210 clusters, including a 2.26× faster training update in a matched four-H100 benchmark with loss and gradient agreement checks.
- **Research platform.** A self-hosted Kubernetes (k3s) platform with TLS and single sign-on serves live training playgrounds, run

review and analysis reports regenerated from committed data.

**Conference presentations**

**"Temporal positionality in conversational AI: why timing is not latency."** Accepted talk, Australian Linguistic Society Annual Conference (ALS 2026), University of Western Australia, Perth, 1–4 December 2026.

**"We are a strange land (κοινός χθών)."** POM: Synthetic Sentience, Politics of the Machines, Perth, 2025. Abstract accepted; not presented.

**"Theological, Astrological and Humoral Theories of the Affections and the Diseases of the Soul."** Perth Medieval and Renaissance Group and UWA Centre for Medieval and Early Modern Studies Annual Conference, October 2019.

**Scholarships and prizes**

- Research Training Program Scholarship, Australian Government, University of Western Australia | 2026–
- Ramsay Postgraduate Scholarship, Ramsay Centre for Western Civilisation | 2021
- Philippa Maddern Memorial Prize in Medieval and Early Modern Studies, University of Western Australia | 2019
- Mannkal European Study Tour | 2020
- Mannkal travel bursaries: Samuel Griffith Society Conference (August 2019), ALS Friedman Conference (May 2019), Liberty Fest (March 2019)
- Study Abroad Scholarship, University of Western Australia to Beijing Foreign Studies University | 2018

**Teaching and curriculum leadership**

**Lecturer, AI and Data Science, North Metropolitan TAFE** | August 2023–present

- Built and leads the AI programme in the Integrated Technologies and Cybersecurity portfolio.
- Developed the ICTSS00120 Artificial Intelligence Skill Set, making North Metropolitan TAFE the first TAFE nationally to deliver it, with ethics and discussion-based learning alongside students training their own models.
- Designed and delivers the Certificate IV in ICT (Data Science and AI) local pathway, registered with the Training Accreditation Council, which attracted a full first cohort.
- Teaches Python, applied machine learning, deep learning, data analysis and the ethics of AI deployment to students and industry professionals from health, mining and research.

**Geometry and the Scientific Method (CUARES403), North Metropolitan TAFE** | 2025. Designed and taught a seminar introduction to the philosophy of science, adapting the St John's College tutorial method to vocational students. Students constructed and defended Euclid's proofs at the board (*Elements* I, Definitions 1–17 and Propositions 1–48) before reading Plato's *Meno*, Descartes' *Discourse on Method*, Lobachevsky's *Theory of Parallels* and Einstein's *Geometry and Experience*, asking what it means for knowledge to be demonstrated.

**Graduate Institute Greek Tutor, St John's College, Annapolis** | 2022–2023. Guided Master's students' Greek translation in one-to-one and small-group tutorials, with line-by-line feedback on grammar, lexicon and interpretation.

**Service and national advisory roles**

- **Member, AI Specialist Technical Committee, Future Skills Organisation** | current. Advises the national ICT training package review on AI qualifications as a subject-matter expert and AI educator, for the Australian Government's Jobs and Skills Council.
- **Technical Committee Member, Future Skills Organisation**: Training Package Project, Qualification Design to Support Digital Capability.
- **Participant, Future Skills Organisation Qualification Reform Demonstration Project**: Digital Capability Uplift.
- **Focus group participant, Future Skills Organisation Finance, Technology and Business VET Workforce Study**, advising on the categorisation of ICT qualifications in national qualifications reform.
- **Committee Member (Technology Services), UWA Historical Society** | 2019–2020. Managed the society's technology projects, including oversight of a student augmented-reality history project.

**Industry and research engagement**

- Secured sponsored access from Hewlett Packard Enterprise to an HPC server with two H100 GPUs, and fine-tuned an 11-billion-parameter Llama 3.2 Vision model on it with parameter-efficient methods.
- Negotiated HPC upskilling placements for lecturing staff with the Curtin Institute of Radio Astronomy and the Australian SKA Regional Centre (AusSRC).
- Partnered with the WA Data Science Innovation Hub on a business case for AI infrastructure investment in Western Australia to support skills training.
- Established links with the Pawsey Supercomputing Centre, UWA and Curtin computing faculties, and represented North Metropolitan TAFE at the CEDA AI Summit 2024.

**Professional experience**

**Founder and Director, Prosody Labs, Perth** | 2024–present. Founded the independent AI research company developing Kairos, and built its agent-native experiment system and research platform.

**Chief Technology Officer, Urban Peak Hour Technologies** | 2023–2025. Led technical strategy for an AI traffic-safety proof of concept, coordinating requirements and data security with the Victorian Department of Transport and Planning, and oversaw the minimum viable product.

**Machine Learning Engineer (Research Assistant), AIMES Laboratory, University of Melbourne** | 2022. Automated deployment with infrastructure as code and Kubernetes, rebuilt data pipelines around batch and event-driven processing, and made

automated training and prediction more reliable with message queueing.

**Software Developer / Solutions Architect, InSig Technologies** | 2021. Developed a mining information system for RUC Mining (now part of the Redpath Group) across the full stack.

**Graduate Software Developer, Risk Management Technologies** | 2020. Developed workplace health and safety risk-management software in Java and AngularJS.

**Communications and Research Intern, Office of Multicultural Interests, Government of Western Australia** | 2019.

**Languages and skills**

**Languages:** Ancient Greek (reading and translation).
**Machine learning:** PyTorch; recurrent, state-space and attention sequence models; audio representation learning; language modelling; distributed training on CUDA and ROCm; numerical validation and gradient checking.
**Engineering:** Python, Triton, JavaScript, Slurm, Kubernetes, Docker, Linux, Git, FastAPI.